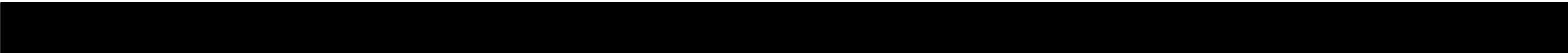


Day 4

Scaling IoT and Edge AI

Hands-on session

A solid black horizontal bar spanning the width of the slide at the bottom.

IoT Model Canvas

<i>Designed by:</i>				<i>On:</i>
				<i>Iteration #</i>
HARDWARE	PROTOCOLS	DATA	APPS / DASHBOARDS	ACTORS
	PAYLOADS		CLOUD CODE	
VALUE PROPOSITION			3rd PARTY INTEGRATIONS	

My classes (4)

~~● Business models and the Internet of Things.~~

~~● Business Model Canvas~~

—

~~● IoT / Edge AI project (hands on)~~

—

~~● The IoT Stack How to build your IoT project, digital transformation and more.~~

~~● IoT Model Canvas~~

—

● Scale your IoT / Edge AI projects (hands-on)

Today

Today

- CI / CD Pipeline
- Let's scale a simple Edge AI object detection project with Raspberry Pi 4 and a Webcam

First question...

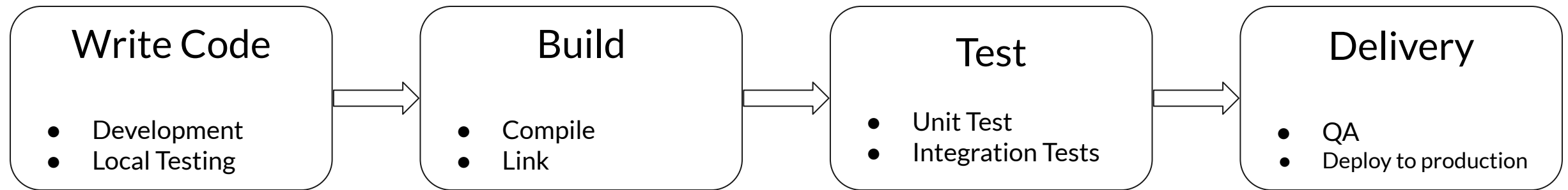
**How to
scale to
the moon?**



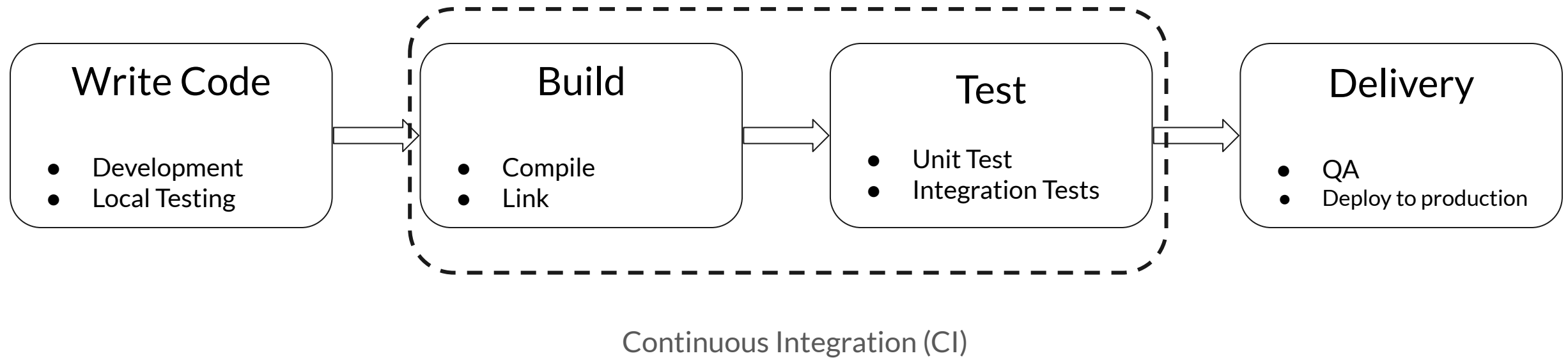
CI / CD Pipeline

Reference: <https://www.youtube.com/watch?v=1nxGcflm-TU&list=PLEBQazB0HUyR00bgock-5uZoOrbmPASC&index=2>

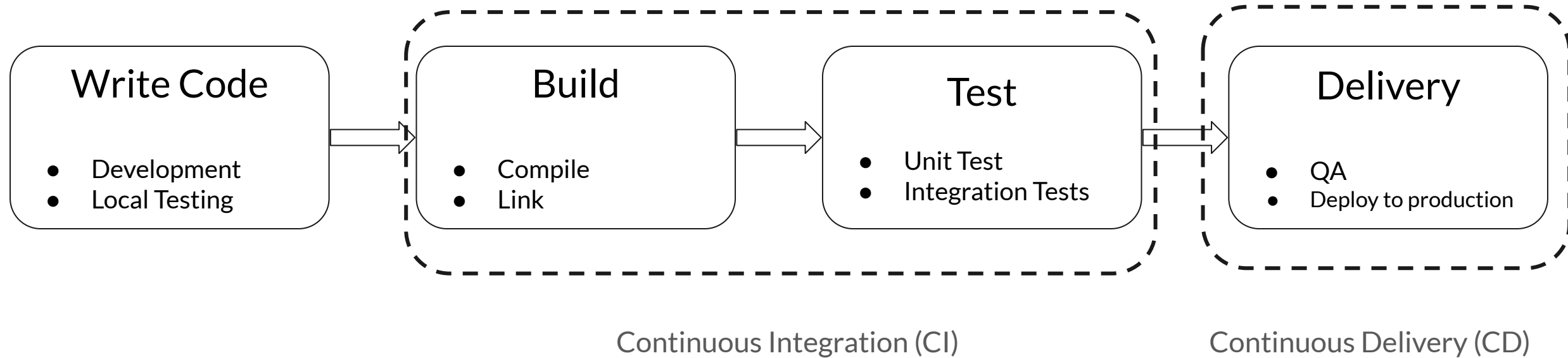
CI / CD Pipeline



CI / CD Pipeline



CI / CD Pipeline





GitHub Actions

Workflow runs · arduino/comp

github.com/arduino/compile-sketches/actions

FeinaReinicia Chrome per fer l'actualització

arduino / compile-sketches

Type to search

CodeIssues11Pull requests4ActionsSecurityInsights

Actions

All workflows

Check Action Metadata

Check Files

Check General Formatting

Check License

Check Markdown

Check npm

Check Poetry

Check Prettier Formatting

Check Python

Check Taskfiles

Show more workflows...

Management

Caches

Attestations

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

7,969 workflow runs

EventStatusBranchActor

✓ Sync Labels

Sync Labels #1260: Scheduled

main

Dec 8, 9:05 AM GMT+1

34s

✓ Sync Labels

Sync Labels #1259: Scheduled

main

Dec 7, 9:04 AM GMT+1

58s

✓ Merge pull request #441 from arduino/dependabot/pip/urllib3-2....

Test Python #1252: Commit b3989dc pushed by per1234

main

Dec 6, 12:03 PM GMT+1

53s

✓ Merge pull request #441 from arduino/dependabot/pip/urllib3-2....

Check Poetry #588: Commit b3989dc pushed by per1234

main

Dec 6, 12:03 PM GMT+1

40s

✓ Merge pull request #441 from arduino/dependabot/pip/urllib3-2....

Check General Formatting #1481: Commit b3989dc pushed by per1234

main

Dec 6, 12:03 PM GMT+1

20s

✓ Merge pull request #441 from arduino/dependabot/pip/urllib3-2....

Check Python #1247: Commit b3989dc pushed by per1234

main

Dec 6, 12:03 PM GMT+1

38s

✓ Merge pull request #441 from arduino/dependabot/pip/urllib3-2....

Run integration tests #1468: Commit b3989dc pushed by per1234

main

Dec 6, 12:03 PM GMT+1

1m 48s

✓ Merge pull request #441 from arduino/dependabot/pip/urllib3-2....

main

Dec 6, 12:03 PM GMT+1

15

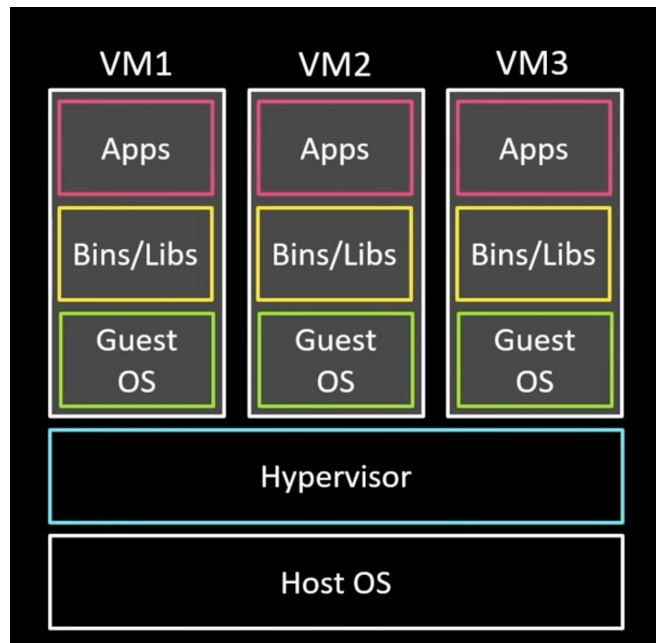
Docker Containers

Docker Containers

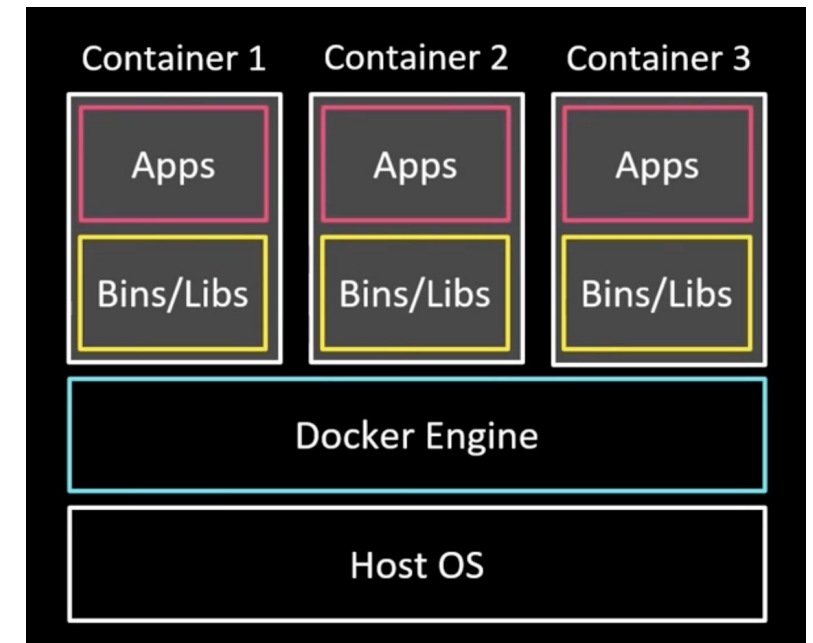
Docker is a platform that lets you run applications inside of virtualized instances know as containers.

Docker Containers

Docker is a platform that lets you run applications inside of virtualized instances know as containers.



Virtual Machines



Docker


```
1 FROM balenalib/raspberrypi5-debian-python:bookworm-build
2
3 # Next three lines courtesy of https://github.com/hyzhak/pi-camera-in-docker/blob/main/Dockerfile
4
5 # Mimic Pi OS:
6 RUN apt update && apt install -y --no-install-recommends gnupg
7
8 RUN echo "deb http://archive.raspberrypi.org/debian/ bookworm main" > /etc/apt/sources.list.d/raspi.list \
9     && apt-key adv --keyserver keyserver.ubuntu.com --recv-keys 82B129927FA3303E
10
11 RUN apt update && apt -y full-upgrade
12
13 # Install rpicas apps prereqs
14
15 RUN apt install -y libcamera-dev libepoxy-dev libjpeg-dev libtiff5-dev libpng-dev \
16     cmake libboost-program-options-dev libdrm-dev libexif-dev meson ninja-build
17
18 WORKDIR /usr/src
19
20 # Clone and build rpicas apps
21
22 RUN git clone https://github.com/raspberrypi/rpicam-apps.git && cd /usr/src/rpicam-apps && \
23     meson setup build -Denable_libav=disabled -Denable_drm=enabled -Denable_egl=disabled -Denable_qt=disabled -Denable_opencv=disabled
24
25 ENV UDEV=on
26
27 COPY *.sh ./
28
29 CMD ["sleep", "infinity"]
```

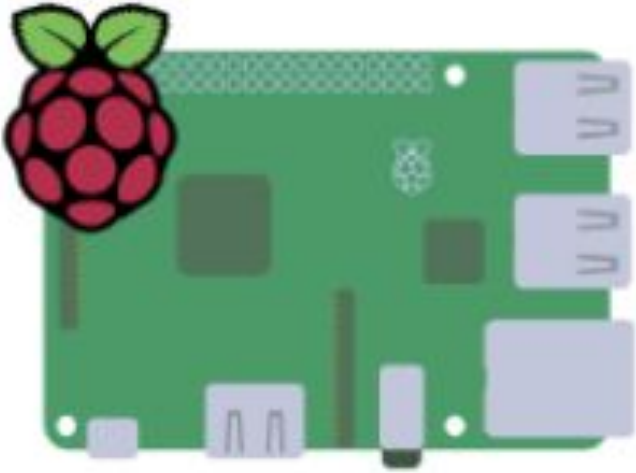
docker-compose.yml

```
1  version: '2'
2
3  services:
4    pi-cam:
5      build: .
6      privileged: true
7      ports:
8        - "80:80"
9      labels:
10       io.balena.features.firmware:
```

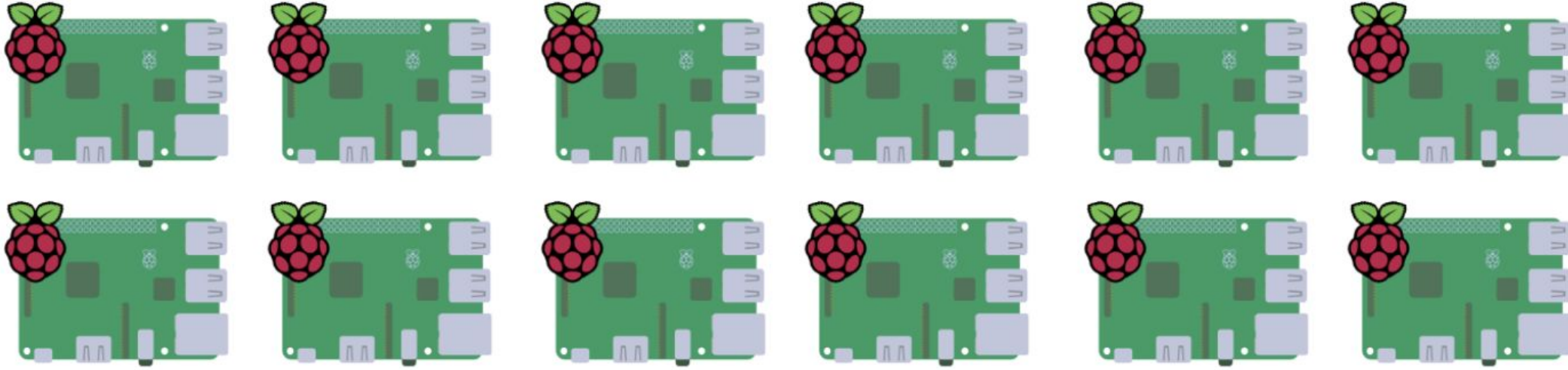
```
57  # -----
58  # MQTT
59  # -----
60  mqtt:
61    image: eclipse-mosquitto:1.6.15
62    container_name: mqtt
63    ports:
64      - "1883:1883"
65    restart: always
66
67  # -----
68  # Influxdb
69  # -----
70  influxdb:
71    image: influxdb:1.7.11
72    container_name: influxdb
73    restart: unless-stopped
74    volumes:
75      - influxdb-data:/var/lib/influxdb
76      - influxdb-etc:/etc/influxdb
77
78  # -----
79  # Grafana
80  # -----
81  grafana:
82    #image: balenablocks/dashboard
83    image: grafana/grafana
84    container_name: grafana
85    restart: unless-stopped
86    ports:
87      - "8080:3000"
88    volumes:
89      - "grafana-data:/data"
```

How to scale?

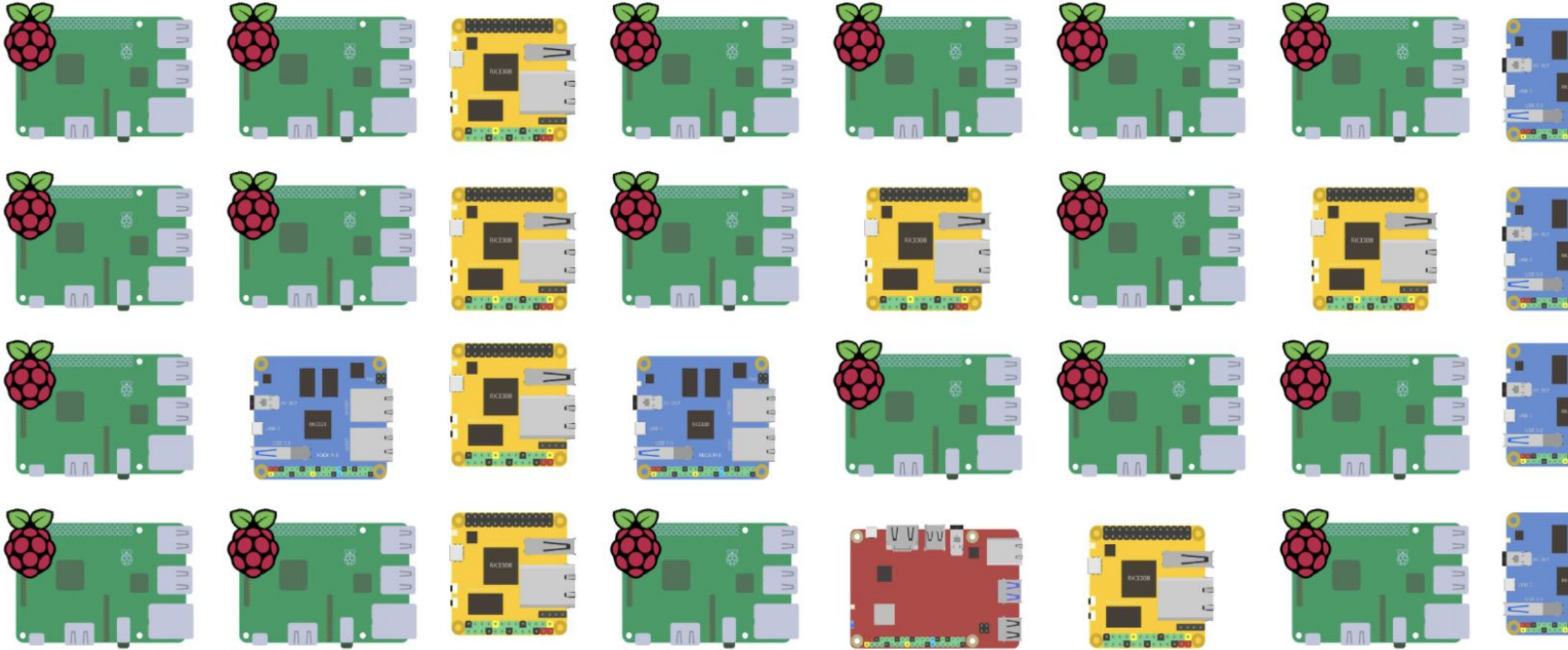
How to scale?



How to scale?



How to scale?



Fleet Management

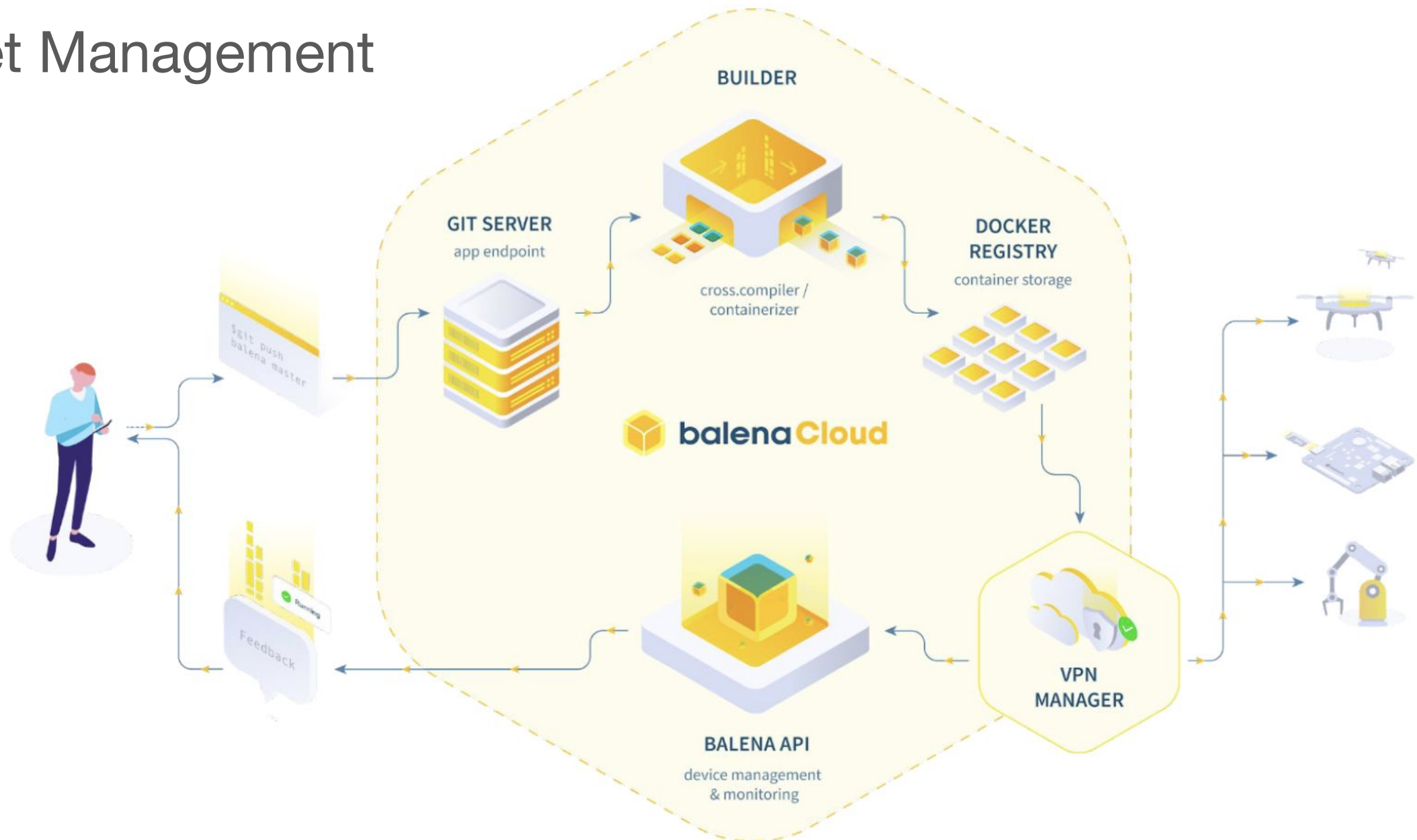
Microcontrollers



Single Board Computers

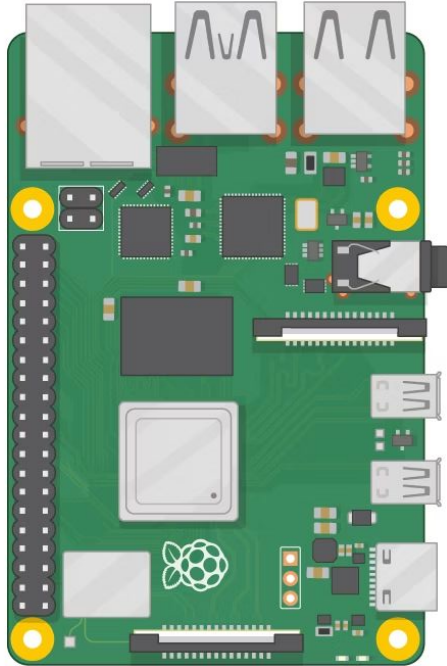


Fleet Management

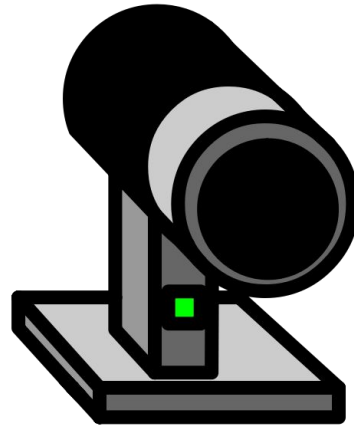


Let's **scale** together a simple Edge AI object detection **fleet** with Raspberry Pis 4 and a Webcams **deployed all around the world.**

Edge AI Project: Object Detection



Raspberry Pi 4



USB Webcam with
microphone



Micro SD card

Create Edge Impulse account

<https://edgeimp.com/master25>



Create a balena account

<https://balena.io>

Download the [balena CLI](#)

The project

The screenshot shows the GitHub repository page for `mpous/balena-cam-mqtt`. The repository is public and has 1 branch (master) and 0 tags. The commit history shows 182 commits, with the latest commit by `mpous` titled "Refactor response handling in server.py" 9 hours ago. The file list includes:

File	Commit Message	Time Ago
<code>.github</code>	Enable deploy to balena action	3 years ago
<code>.versionbot</code>	v0.2.10	3 years ago
<code>balena-cam-mqtt</code>	Refactor response handling in server.py	9 hours ago
<code>balena-cam</code>	Improving the code: Pin pyee to version 8.2.2	3 years ago
<code>.gitignore</code>	Dockerfile: Add Dockerfile, basic app structure, .gitignore	7 years ago
<code>CHANGELOG.md</code>	v0.2.10	3 years ago
<code>CONTRIBUTING.md</code>	patch: add CONTRIBUTING.md	3 years ago
<code>LICENSE</code>	Create LICENSE	3 years ago
<code>README.md</code>	Update README.md	last year
<code>VERSION</code>	v0.2.10	3 years ago
<code>balena.yml</code>	v0.2.10	3 years ago
<code>docker-compose.yml</code>	Comment out network_mode in docker-compose.yml	10 hours ago
<code>logo.png</code>	meta: update balena.yml	4 years ago
<code>repo.yml</code>	repo: Add repo.yml	5 years ago

The right sidebar contains sections for "About" (No description, website, or topics provided), "Releases" (No releases published), "Packages" (No packages published), "Contributors" (13 contributors), and "Languages" (Python 36.2%, JavaScript 28.9%).

<https://github.com/mpous/balena-cam-mqtt/tree/master>

git clone [git@github.com:mpous/balena-cam-mqtt.git](https://github.com/mpous/balena-cam-mqtt.git)

balena login

balena push <org-name/fleet-name>

```
[balena-cam-mqtt] Removing intermediate container 930133c7c929
[balena-cam-mqtt] ---> 4db92c53a290
[balena-cam-mqtt] Step 8/10 : WORKDIR /usr/src/app
[balena-cam-mqtt] ---> Running in 7d49d4f66ed0
[balena-cam-mqtt] Removing intermediate container 7d49d4f66ed0
[balena-cam-mqtt] ---> b27ffc2e49d9
[balena-cam-mqtt] Step 9/10 : COPY ./app/ /usr/src/app/
[balena-cam-mqtt] ---> 022dec5c7440
[balena-cam-mqtt] Step 10/10 : CMD ["python3", "/usr/src/app/server.py"]
[balena-cam-mqtt] ---> Running in 8f935418d16e
[balena-cam-mqtt] Removing intermediate container 8f935418d16e
[balena-cam-mqtt] ---> 0ecd2ea7e5df
[balena-cam-mqtt] Successfully built 0ecd2ea7e5df
[Info] Uploading images
[Success] Successfully uploaded images
[Info] Built on 28fcfdc
[Success] Release successfully created!
[Info] Release: 4f862f7e1cf5a657c730cda74d77f6bd (id: 3799706)
[Info]
```

Service	Image Size	Build Time
balena-cam-mqtt	2.19 GB	1 minute, 57 seconds

```
[Info] Build finished in 4 minutes, 25 seconds
```

marcpous@Marc-Pous-GJQJX9NM7X balena-cam-mqtt %

balenaCam-mqtt | balena da

dashboard.balena-cloud.com/fleets/2318836/summary

FeinaReinicia Chrome per fer l'actualització

balenaCloudIoT Dashboard

Organizations

g_marc_pous2's Organi...

Fleets

balena-cam-mqtt

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balena-cam-mqtt

ARCHITECTURE

aarch64

SLUG

g_marc_pous2/balena-cam-mqtt

CREATED

Dec 9th 2025, 12:43 AM

Devices

0

You don't have any devices yet.

How about adding one?

+ Add device

Releases

1

track latest

balena-cam

Learn more

Create release

This is your fleet summary.

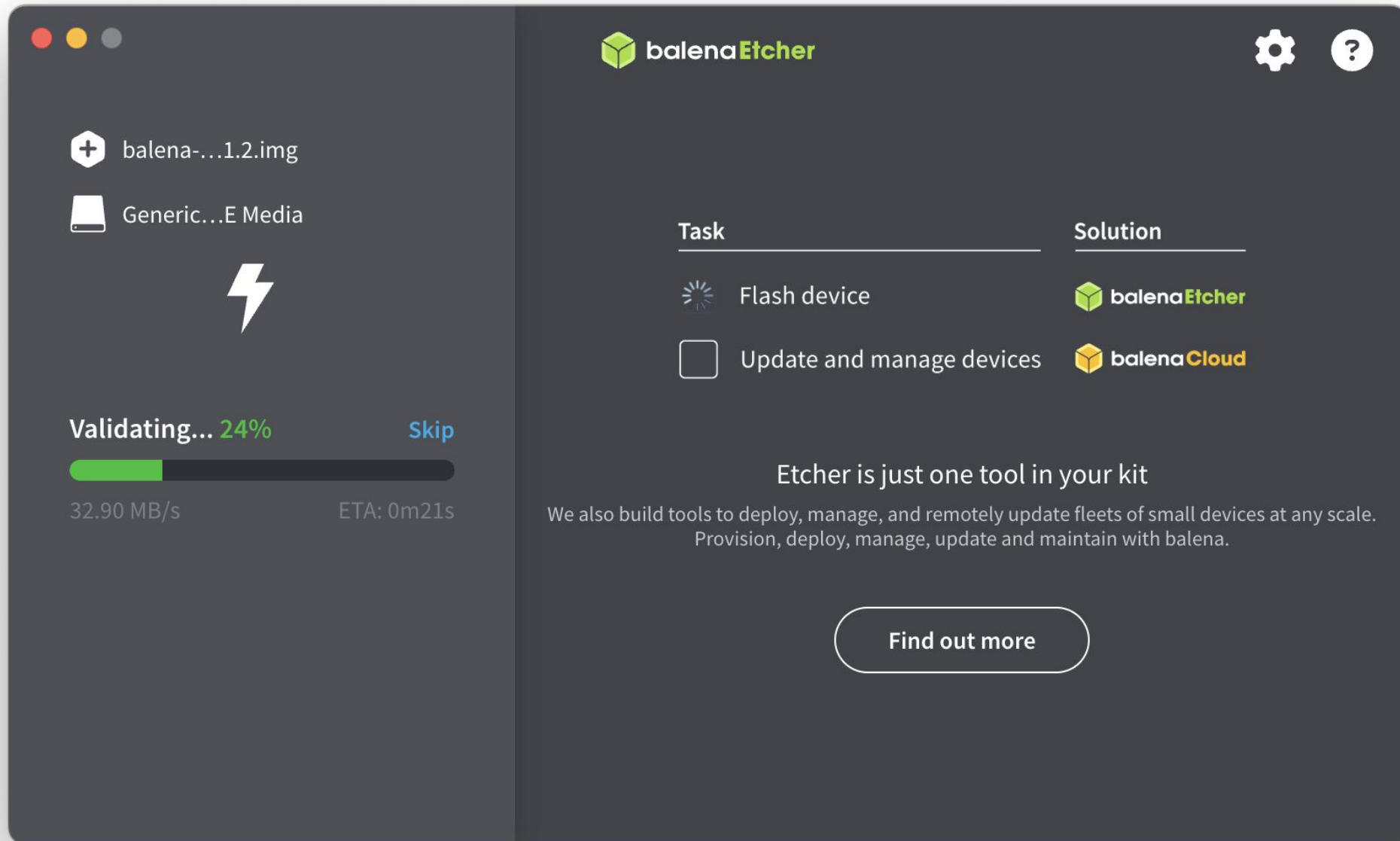
It's a bit empty right now, looks like you need some devices. Let's go ahead and add one.

+ Add device

Need help

Changelogv31.4.26

33



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balena-cam-mqtt



ARCHITECTURE
aarch64
SLUG
g_marc_pous2/balena-cam-mqtt
CREATED
Dec 9th 2025, 12:43 AM

Devices

1

operational

Releases

1

track latest

balena-cam

Learn more

Create release

Add device

Modify

Search

Add filter

Views

Manage columns

☐	Name ↑	UUID	Status	Heartbeat	Cloudlink	Cloudlink last connected	Current release	Target release	OS version	IP address	Tag
☐	sleek-snow	ed9dfd4	Operational	✓	✓	Connected (for 8 minutes)	0.2.10+rev1 ✓	0.2.10+rev1	balenaOS 6.7.7+rev1	192.168.1.136	

Rows per page: 50 1-1 of 1

Need help

Changelog
v31.4.26

The screenshot shows a web browser window with the URL `docs.edgeimpulse.com/tutorials/topics/lifecycle-management/ota-balena`. The page is titled "Balena" and is part of the "Lifecycle management" tutorial series. The left sidebar contains a navigation menu with categories like Machine learning, Inference, Post-processing, Lifecycle management, and Balena. The main content area has a breadcrumb trail: TOPICS > Lifecycle management > Perform OTA model updates. The page is divided into three sections: Introduction, Prerequisites, and Overview. The Introduction section explains that the page is part of the Lifecycle management with Edge Impulse tutorial series and recommends reading the introduction first. The Prerequisites section lists three requirements: an active Edge Impulse account, following the Edge Impulse Docker documentation, and having a Balena Example Repo. The Overview section states that Balena offers a comprehensive platform for building, deploying, and managing large fleets of IoT devices. The right sidebar contains a table of contents for the page, listing sections like Introduction, Prerequisites, Overview, and Conclusion.

EDGE IMPULSE Search... Website Login Sign up >

Knowledge Studio Hardware Tools APIs **Tutorials** Projects Datasets

Machine learning >
Inference >
Post-processing >
Lifecycle management >
Acquire and process data using Golioth on AI
Deploy using GitHub Actions
Perform OTA model updates >
Allxon
Arduino IDE (for ESP32)
Arduino IoT Cloud
Balena
Blues Wireless
C++
Espressif IDF
Nordic Thingy53
NVIDIA Jetson
Particle Workbench
Particle Webhook
Zephyr on Golioth
Zephyr >
Android >

TOPICS > Lifecycle management > Perform OTA model updates

Balena

Introduction

This page is part of the Lifecycle management with Edge Impulse tutorial series. If you haven't read the introduction yet, we recommend you to do so [here](#). Balena can serve as the platform for deploying OTA updates to edge devices, including new models trained with Edge Impulse.

Prerequisites

- An active Edge Impulse account with a project and trained impulse. See the [Edge Impulse - Object Detection Tutorials](#) for more information on how to create a project and train an impulse.
- Follow the [Edge Impulse Docker documentation](#).
- [Balena Example Repo](#) - A sample balena project to deploy your impulse as on balena. It exposes an API which the "cam" service uses to get repeated inferences from the images captured by the webcam. See the [README](#)

Overview

Balena offers a comprehensive platform for building, deploying and managing large fleets of IoT devices. It simplifies fleet management, enhances security, and

On this page

- [Introduction](#)
- [Prerequisites](#)
- [Overview](#)
- [Prerequisites](#)
- [Introduction to balena](#)
 - Step 1: Exporting Your Model as a Docker Container
 - Step 2: Preparing Your Balena Fleet
 - Step 3: Deploying Your Docker Container to Your balena Fleet
 - Step 4: Build Your Application
 - Step 5: Accessing Your Inference Server
 - Step 6: Monitoring and Managing Your Fleet
- [Conclusion](#)

<https://docs.edgeimpulse.com/tutorials/topics/lifecycle-management/ota-balena>

Object Detection - Rubber Duckies

studio.edgeimpulse.com/studio/783798/acquisition/training?page=1

Feina

Finalitza l'actualització

EDGE IMPULSE

Dashboard

Devices

Data acquisition

Experiments

EON Tuner

Impulse design

Create impulse

Image

Object detection

Retrain model

Live classification

Model testing

Post-processing

Deployment

Versioning

GETTING STARTED

Marc using Arduino UNO Q / Object Detection - Rubber Duckies

ENTERPRISE

Target: Raspberry Pi 5

Dataset

Data sources

Synthetic data

Labeling queue (0)

AI labeling

DATA COLLECTED

212 items

TRAIN / TEST SPLIT

82% / 18%

Dataset

Training (168)

Test (41)

Post-processing (3)

unknown.67qkjj9n

unknown.67qkf6ai

unknown.67qkera4

unknown.67qkefq9

Captura de pantalla 20:

Picture.64rn5jhc

Picture.64rn6l0s

Picture.64rn6suk

Picture.64rn77un

Collect data

Device

Pi 4

Name

sample

Sensor

Camera (640x480)

Camera feed

Start sampling

Click on a sample to load...

Object Detection - Rubber Duckies

studio.edgeimpulse.com/studio/783798/impulse/1/deployment

Feina

Reinicia Chrome per fer l'actualització

EDGE IMPULSE

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EON Tuner

Impulse design

Create impulse

Image

Object detection

Retrain model

Live classification

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GETTING STARTED

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Forums

Discord

Marc using Arduino UNO Q / Object Detection - Rubber Duckies

ENTERPRISE

Target: Raspberry Pi 5

Configure your deployment

You can deploy your impulse to any device. This makes the model run without an internet connection, minimizes latency, and runs with minimal power consumption. [Read more.](#)

Q Docker container

SELECTED DEPLOYMENT

Docker container

A Docker container with an HTTP interface to run your models.

RUN YOUR MODEL AS A DOCKER CONTAINER

To run your model as a container with an HTTP interface, use:

Container:
`public.ecr.aws/g7a8t7v6/inference-container:v1.80.0`

This automatically builds and downloads the latest model (incl. hardware optimizations), and runs an HTTP endpoint at <http://localhost:1337> with instructions.

[Read the docs](#) for information, like bundling in your model inside the container and selecting extra hardware optimizations.

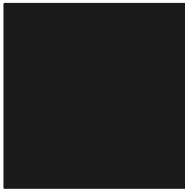
Latest build

 v5 (Linux (AARCH64))
nov. 25 2025, 15:31:21

View docs

Run this model

Scan QR code or launch in browser to test your prototype



Launch in browser

?

38

40

sleek-snow | balena dashboard

Object Detection - Rubber Duck

Marc using Arduino UNO R3

dashboard.balena-cloud.com/devices/ed9dfd41a5bb964e8b5b9ab8973278bd/summary

Feina

Reinicia Chrome per fer l'actualització

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sleek-snow

Actions

FLEET

g_marc_pous2/balena-cam-mqtt

STATUS

Operational

HEART BEAT

UPDATE STATUS

Done

OS VARIANT

DEVELOPMENT

TARGET RELEASE

0.2.10+rev2

LOCAL MODE

LOCAL IP ADDRESS

192.168.1.136

MAC ADDRESS

DC:A6:32:B1:07:42

DC:A6:32:B1:07:43

TAGS (0)

No tags configured yet

NOTES

No notes

CLOUDLINK

UUID

ed9dfd4

UPDATE STATUS DURATION

1 minute

SUPERVISOR VERSION

17.1.2

SUPPORT ACCESS

Off

PUBLIC DEVICE URL

PUBLIC IP ADDRESS

46.6.127.174

CLOUDLINK LAST CONNECTED

Connected (for 27 minutes)

TYPE

Raspberry Pi 4 (using 64bit OS)

HOST OS VERSION

balenaOS 6.7.7+rev1 (Changelog)

CURRENT RELEASE

0.2.10+rev2

IS ACTIVE

CPU

54%

Temperature

55C

Memory

555 MB/7.7 GB

Storage

(/dev/mmcblk0p6) 3.3 GB/28.3 GB

Logs

UTC

Timestamps

Add filter

Views

balena-cam-mqtt

192.168.1.136

balena-cam-mqtt

1883

balena-cam-mqtt

/balena

balena-cam-mqtt

python-mqtt-1765240860

balena-cam-mqtt

192.168.1.136

balena-cam-mqtt

1883

balena-cam-mqtt

/balena

mqtt

1765240860: New connection from 172.17.0.1 on port 1883.

balena-cam-mqtt

Sent image to topic '/balena'

mqtt

1765240860: New client connected from 172.17.0.1 as python-mqtt-1765240860 (p2, c1, k60).

balena-cam-mqtt

Connected to MQTT Broker!

mqtt

1765240860: Socket error on client python-mqtt-1765240860, disconnecting.

Terminal

Select a target

Start terminal session

Changelog v31.4.26

41

```
[edge-impulse] Successfully built 37a8e2e88acd
[influxdb] [=====] 100%
[node-red] [=====] 100%
[Info] Generating image deltas from release b5e91daa9d80a6a024388d3057de4415 (id: 3799757)
[Success] Successfully generated image deltas
[Info] Uploading images
[Success] Successfully uploaded images
[Info] Built on 28fcfdc
[Success] Release successfully created!
[Info] Release: 214a2336fdcf0d804d0f56858aae2ec3 (id: 3799764)
```

Service	Image Size	Delta Size	Build Time
balena-cam-mqtt	2.19 GB	0 bytes	< 1 second
edge-impulse	767.23 MB	0 bytes	1 second
mqtt	10.71 MB	0 bytes	< 1 second
node-red	525.82 MB		33 seconds
influxdb	264.99 MB		10 seconds

Build finished in 1 minute, 41 seconds



balenaCloud

IoT Dashboard

Organizations

g_marc_pous2's Organi...

Fleets

balena-cam-mqtt

Devices

sleek-snow

Summary

Device Variables

Device Configuration

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Diagnostics

Location

UPDATE STATUS

Done

OS VARIANT

DEVELOPMENT

TARGET RELEASE

0.2.10+rev3

LOCAL MODE

LOCAL IP ADDRESS

192.168.1.136

MAC ADDRESS

DC:A6:32:B1:07:42

DC:A6:32:B1:07:43

TAGS (0)

No tags configured yet

NOTES

No notes

ed9dfd4

UPDATE STATUS DURATION

less than a minute

SUPERVISOR VERSION

17.1.2

SUPPORT ACCESS

Off

PUBLIC DEVICE URL

PUBLIC IP ADDRESS

46.6.127.174

Raspberry Pi 4 (using 64bit OS)

HOST OS VERSION

balenaOS 6.7.7+rev1 (Changelog)

CURRENT RELEASE

0.2.10+rev3

IS ACTIVE

Getting Started

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UTC

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balena-cam-mqtt

Return the image as a bytes object

balena-cam-mqtt

Send MQTT image...

balena-cam-mqtt

192.168.1.136

balena-cam-mqtt

1883

balena-cam-mqtt

/balena

balena-cam-mqtt

python-mqtt-1765241324

balena-cam-mqtt

192.168.1.136

balena-cam-mqtt

1883

balena-cam-mqtt

/balena

mqtt

1765241324: New connection from 172.17.0.1 on port 1883.

mqtt

1765241324: New client connected from 172.17.0.1 as python-mqtt-1765241324 (p2, c1, k60).

balena-cam-mqtt

Sent image to topic '/balena'

balena-cam-mqtt

Connected to MQTT Broker!

mqtt

1765241324: Socket error on client python-mqtt-1765241324, disconnecting.

Terminal

Select a target

Start terminal session

Services

Status

Release

balena-cam-mqtt

Running

0.2.10+rev3

edge-impulse

Running

0.2.10+rev3

influxdb

Running

0.2.10+rev3

mqtt

Running

0.2.10+rev3

node-red

Running

0.2.10+rev3

Changelog

v31.4.26

Balena Cloud

- Organizations
- Fleets
- balena-cam-mqtt
- Devices
- sleek-snow
 - Summary
 - Device Variables**
 - Device Configuration
 - Actions
 - Settings
 - Diagnostics
 - Location

Name ↑	Fleet value	Device value	Service name	Actions
mqttn_broker	not defined	192.168.1.136	All services	[Edit] [Delete]
mqttn_port	not defined	1883	All services	[Edit] [Delete]
mqttn_topic	not defined	/balena	All services	[Edit] [Delete]
PASSWORD	not defined	balena	All services	[Edit] [Delete]
USERNAME	not defined	balena	All services	[Edit] [Delete]

Rows per page: 50 | 1-5 of 5

Browser tabs: sleek-snow | balena dashboard, Object Detection - Rubber Duck, Marc using Arduino UNO Q / C, balena Node-RED : 192.168.1.1

Address bar: No segur 192.168.1.136/#flow/2669f04f8fb1c912

balena Node-RED

filter nodes

Flow 1

common

- inject
- debug
- complete
- catch
- status
- link in
- link call
- link out
- comment

function

- function
- switch
- change
- range
- template
- delay
- trigger

info

Search flows

Flows

- Flow 1
- Subflows
- Global Configuration Nodes

Flow 1

Flow "2669f04f8fb1c912"

1/9

Start >

Enable or disable these tips from the option in the settings

sleek-snow | balena dashboard x Object Detection - Rubber Duck x Marc using Arduino UNO Q / x balena Node-RED : 192.168.1.1 x

No segur 192.168.1.136/#flow/2669f04f8fb1c912

Feina Reinicia Chrome per fer l'actualització

balena Node-RED

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debug 1

connected

debug

all nodes

all

9/12/2025 1:52:54 node: debug 1

/balena : msg.payload : string[102352]

" /9j /4AAQSkZJRgABAQAAQABAAZ /2wBDAAI
BAQEBAQIBAQECAGICAgQDAgICAgUEBAMEBgU
GBgYFBgYGBwkIBgcJBWYGCAsICQoKCgoKBgg
LDAsKDAkKCgr /2wBDAQICAgICAgUDAwUKBwY
HCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgo
KCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgr /wAA
RCAHgAoADASIAAhEBAxEB /8QAHwAAAQUBAQE
BAQEAAAAAAAAAAAECAwQFBgcICQoL /8QAtRA
AAgEDAwIEAwUFBAQAAAF9AQIDAAQRBRIhMUE
GE1FhByJxFDKBkaEII0KxwRVS0fAkM2Jyggk
KFhcyGRoLJicoKSo0NTY3ODk6Q0RFRkdISUp
TVFVWV1hZWmNkZWZnaGlqc3R1dnd4eXqDhIW
Gh4iJipKTlJWWl5iZmqKjpKWmp6ipqrKztLW
2t7i5usLDxMXGx8jJyLTlINXW19jZ2uHi4+T
l5ufo6erx8vP09fb3+Pn6 /8QAHwEAAwEBAQE
BAQEBAQAAAAAAAAECAwQFBgcICQoL /8QAtRE
AAgECBAQDBAcFBAQAAQJ3AAECAxEEBSExBhJ
BUQdhcRMiMoEIFEKRobHBCSMzUvAVYnLRChY
kNOEl8RcYGRomJygpKjU2Nzg50kNERUZHSEl
KU1RVVldYWVpjZGVmZ2hpanN0dXZ3eHl6go0
EhYaHiImKkpOUlZaXmJmaoq0kpaanqKmqsr0
0tba3uLm6wsPExcbHyMnK0tPU1dbX2Nna4uP
k5ebn60nq8vP09fb3+Pn6 /9oADAMBAAIRAxE
APwD6DIY965v4hfB74dfftS30nxz4Xtb6N1I
35xgsPoa6ZRNbQiv8vKGIrYaoqLKTjJbNaM
/pOUYyVmro+Fv2g /+CNngzxEO2vfbZvzp10c
strKfLPsK+J /jt+xn8d /hBJLYeMfA01xDESI
722ij49QRX7ggk8darax4f0XxDZtY..."

sleek-snow | balena dashboard x Object Detection - Rubber Duck x Marc using Arduino UNO Q / x balena Node-RED : 192.168.1.1 x

No segur 192.168.1.136/#flow/2669f04f8fb1c912

Feina Reinicia Chrome per fer l'actualització

balena Node-RED

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9/12/2025 1:52:54 node: debug 1

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" /9j /4AAQSkZJRgABAQAAQABAAQ /2wBDAAI
BAQEBAQIBAQECAGICAgQDAgICAgUEBAMEBgU
GBgYFBgYGBwkIBgcJBWYGCAsICQoKCgoKBgg
LDAsKDAkKCgr /2wBDAQICAgICAgUDAwUKBwY
HCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgo
KCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgoKCgr /wAA
RCAHgAoADASIAAhEBAxEB /8QAHwAAAQUBAQE
BAQEAAAAAAAAAAAECAwQFBgcICQoL /8QAtRA
AAgEDAwIEAwUFBAQAAAF9AQIDAAQRBRIhMUE
GE1FhByJxFDKBkaEII0KxwRVS0fAkM2Jyggk
KFhcyGRoLJicoKSo0NTY3ODk6Q0RFRkdISUp
TVFVWV1hZWmNkZWZnaGlqc3R1dnd4eXqDhIW
Gh4iJipKTlJWWl5iZmqKjpKWmp6ipqrKztLW
2t7i5usLDxMXGx8jJyLTlINXW19jZ2uHi4+T
l5ufo6erx8vP09fb3+Pn6 /8QAHwEAAwEBAQE
BAQEBAQAAAAAAAAECAwQFBgcICQoL /8QAtRE
AAgECBAQDBAcFBAQAAQJ3AAECAxEEBSExBhJ
BUQdhcRMiMoEIFEKRobHBCSMzUvAVYnLRChY
kNOEl8RcYGRomJygpKjU2Nzg5OkNERUZHSEl
KU1RVVldYWVpjZGVmZ2hpanN0dXZ3eHl6go0
EhYaHiImKkpOUlZaXmJmaoqOkpaanqKmqsr0
0tba3uLm6wsPExcbHyMnK0tPU1dbX2Nna4uP
k5ebn60nq8vP09fb3+Pn6 /9oADAMBAAIRAxE
APwD6DIY965v4hfB74dfftS30nxz4Xtb6N1I
35xgsPoa6ZRNbQiv8vKGIrYaoqLKTjJbNaM
/pOUYyVmro+Fv2g /+CNngzxEO2vFBzVzp10c
strKfLPsK+J /jt+xn8d /hBJLYeMfA01xDESI
722ij49QRX7ggk8darax4f0XxDZtY..."

More exercises

- Let's build a Telegram bot with Node-RED that send us a notification when a rubber duck is detected?
- And store the events in the InfluxDB database.

My classes (4)

- ~~Business models and the Internet of Things.~~

- ~~Business Model Canvas~~

—

- ~~IoT / Edge AI project (hands on)~~

—

- ~~The IoT Stack How to build your IoT project, digital transformation and more.~~

- ~~IoT Model Canvas~~

—

- ~~Scale your IoT / Edge AI projects (hands on)~~

My Takeaways

- Business models and the Internet of Things.
- It's easy to build a project on the desk → IoT / Edge AI project (hands-on)
- The IoT Stack → Ask the right questions before you start a project.
- Use the IoT Model Canvas
- Buy VS Build
- Remember if you are successful you will need to scale your IoT / Edge AI project

EOF